

Prateek Mehta

Fullstack Software Engineer

Experienced web/mobile/backend and 3D/rendering software developer, adept at fullstack development, test-driven approaches, database and API design. Driving business value through innovation, and proficiency in cutting-edge technologies. Looking for software development roles for best utilization of abilities to build useful products and services for customers.

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Summary

As a fullstack web/mobile/backend and 3D/rendering software developer, the candidate's journey spans multiple roles and impactful contributions. At *Mimik Technology*, they are performing fullstack software development duties using cloud-native patterns, utilizing technologies such as iOS, Android, Node.js, Docker and Kubernetes. They are extending software meant for edge computing, while adapting to various single-board computers for inference with language models, or for developing microservices.

At *Starline Windows*, they developed C# based applications for Autodesk suite of software such as Revit/Inventor, providing business value through automatic report generation for model-coordination and model-build, achieving reduction in shop-errors. At *Manage Petro*, they supported an agile team with fullstack cross-platform mobile development efforts with React Native, ensuring a seamless app architecture, while keeping mobile app project up to date with latest packages.

At *ASEBP*, they orchestrated the entire mobile application development lifecycle, achieving robustness with 90% code coverage through test-driven development. Their consultative approach streamlined requirements-gathering, database and API design. At *Autodesk*, their React and WebXR proficiency powered frontend tools for immersive experiences, while active participation in OpenXR meetings showcased their commitment to cutting-edge technologies.

Integrating the "CarryOn-AR" framework at *Expedia* led to a 3.1% increase in conversions, and their hands-on mobile development skills enhanced trip features for iOS and Android. Additionally, their tenure at *Rolocule* involved creating a unique input library using miracast and motion sensors, and work on network resilience. At *Smartivity*, they launched AR apps, emphasizing attention to detail and frontend excellence. Finally, at *MetaDesign Solutions*, they iteratively improved software applications, reduced project delivery times, and thrived in an agile environment. Their expertise across frameworks, cross-platform ecosystems, and 3D/rendering technologies distinguishes them as a versatile fullstack developer.

Technologies/Tools

Prateek is proficient in various technologies, tools, platforms, and has strong fundamental knowledge, with desire to learn and grow more in specific areas as required by need basis for professional development.

- **Programming/Technologies:** Swift, Swift Package, SwiftUI, RxSwift, Core Data, Realm, Kotlin, Java, Android Library, Jetpack Compose, RxJava, Dagger-Hilt, OkHttp, Retrofit, Jitpack, C#, ASP.NET Core, Python, Django, DRF, JavaScript, Express.js, React, JSX, React Native, Axios, Redux Toolkit, MUI, HTML/CSS, PHP/MySQL, XML/JSON, GraphQL/Apollo, OpenGL ES, Unity/Unreal, Pandas, Matplotlib, RabbitMQ/Celery, Google Cloud/GCP, Azure, AWS, PostgreSQL, MongoDB, Geodatabase
- **Tools:** Git, Docker, Kubernetes, Bash, Android Studio, Xcode, Visual Studio, Gradle, CocoaPods, Unity, .NET, Node.js, NPM/Yarn, Webpack, Expo, StarUML, JupyterLab, Revit/Inventor, ArcGIS, Postman, Figma, Jira, Trello, Pbxproj, Mermaid, Confluence
- **Platforms:** Windows, Linux, macOS, Ubuntu, iOS, Android
- **Fundamentals:** OOP and Design Patterns, Software Architecture, 3D Rendering, Source Control, Data Structures, RDBMS, NoSQL, Reactive Programming, Dependency Injection, CI/CD, Cloud-Native, Introductory Statistics, Remote Sensing, AR/VR/MR, AEC, Design & Make, GIS Analysis, Geodatabase

Professional Record

Prateek has more than nine years of professional software development experience. He has worked for various organizations, ranging from startups to multinational corporations. In most of these roles, he has worn multiple hats and contributed on both full-time and contract bases. He has lived in various provinces, including Ontario, Quebec, British Columbia, and has always been open to relocation for the right opportunity. In his recent roles at *Mimik Technology* and *Starline Windows*, he worked in a hybrid model, with periodic on-site engagement at British Columbia locations.

He is currently seeking hybrid opportunities that allow him to balance professional growth with family responsibilities, and is eager to contribute to forward-thinking teams in both on-site and distributed environments.

Senior Fullstack Developer (Full-Time)

Mimik Technology (From: February 2025, To: Present)

Performed duties for cloud-native software development for mobile and Node.js backend. Extended native iOS/Android apps with UI, API, and IO enhancements, leveraging Realm and custom caching layers. Upgraded mobile apps using modern language features such as AsyncStream for iOS and Flow for Android to support asynchronous data streams within structured concurrency models. Delivered responsive frontend tools with Angular and React, integrated TensorFlow.js for in-browser ML. Improved app performance using platform-specific profiling tools, while adding integration and load tests with XCTest, JUnit, and Mocha. Implemented backend microservices in Node.js with model-processor-controller architecture and Swagger 3.0. Worked across diverse databases and messaging systems, including MongoDB, SQLite, Redis, Firestore, etcd, RabbitMQ. Led development for asynchronous data streams, caching, and backpressure handling, while enhancing offline-first integration testing. Supported infrastructure engineering teams with integration testing on single-board computers, authored CLI automation tools, and managed device/server software, firewalls, load balancers, and containerized services.

Technologies/Tools: Android Studio, Xcode, Angular/React, TensorFlow.js, Realm, WebSocket, Node.js, Swagger 3.0, NGINX, MongoDB, SQLite, Redis, Firestore, etcd, RabbitMQ, XCTest, JUnit, Mocha, Retrofit, GGUF/ONNX, HuggingFace, Service Discovery, OAuth 2.0, Kubernetes, Docker

Key Achievements:

- Suggested IO enhancements to caching layers using in-house storage which improved write operations by 15%, provided chunked streaming documentation for developer understanding to explain use-case driven storage architecture
- Utilized dynamic request body schema using OpenAPI 3.0 for a cleaner architecture, which improved developer onboarding on similar API development tasks
- Participation in company hackathons helped to improve the documentation efforts on external developer documentations, and the overall sitemap to make it convenient for finding useful resources
- Led revamp of core in-house OAuth 2.0 libraries, adding support for standard and on-premise grant types
- Supported infrastructure engineering with integration testing on Raspberry Pi/NVIDIA devices and authored CLI automation tools
- Authored microservices using Node.js, crafted container images using Docker, and operated Kubernetes clusters — streamlining platform reliability across diverse environments
- Development across Ubuntu, OpenWrt, Alpine Linux, macOS, and Windows platforms, helped improve the state of integration testing documentation

Instructor — Fullstack Developer/Data Engineer (Part-Time) **Greystone College (From: January 2024, To: February 2025)**

Worked as an instructor to teach fullstack development for web/mobile platforms using Node.js, Angular, React/React Native. Imparting data engineering concepts such as feature-engineering, data-transformation using JavaScript and Python. This part-time undertaking is to fulfil leadership skills in teaching and software development, and to participate in an environment of emerging professionals coming from different technical backgrounds to blend their expertise into the field of software development.

- Teaching curriculum topics using Moodle, giving live coding demos, explaining the purpose behind creation of software libraries, frameworks, and CLI tools
- Onboarding students to the software development process using tools such as VS Code, Git, and CLI for respective technologies such as Node.js, Angular, React, etc.
- Preparing sample projects and sharing with students, grading their project submissions, providing encouraging feedback
- **Technologies/Tools:** JavaScript/TypeScript, Node.js, Angular, React, Webpack, React Native, Pandas, Firebase Auth, Firestore, MongoDB

Lead Software Engineer (Contract) **Starline Windows (From: July 2024, To: December 2024)**

Collaborated to existing .NET add-ins and created new add-ins while working under the special projects initiative for reducing manual model review efforts by creating tools to suggest and correct modelling errors in Revit and Inventor. Model coordination was facilitated using “validation” add-ins for Revit/Inventor which used Excel-based product rules for model validation. These add-ins reduced shop-errors & provided categorical error-reporting by model parts. Inventor add-in operates in-app & headless (for job queue), while Revit add-in operates in-app only. Also created new projects for generating fab sheets from floor plans to display various window tags with parent frame thumbnails, all using Revit API features.

Technologies/Tools: .NET, WinForms, Revit, Inventor, Vault, Git, GitLab, SQLite, SQL, PowerBI

Key Achievements:

- Standardized modelling workflow with validator ensured accurate error-reporting of incorrect model parameters (e.g. for dimensions), and incorrect model families (e.g. for profile-codes)
- Validator add-ins reduced overall shop-errors by almost 90% by improving upstream modelling workflows
- Categorical error-reporting fed into PowerBI dashboards for advanced monthly analytics helped discovering useful trends
- Fab Maker add-in simplified sheet generation, and reduced dependence on an obsolete third-party tool
- It also reduced sheet generation time down to a few seconds from many hours
- Candidate also worked with teams from SolidCAD to improve Vault workflows for faster searching of Inventor files, and extension of PLM lifecycle in Vault

Lead Software Engineer (Contract) **Manage Petro (From: February 2024, To: May 2024)**

Integrating frontend components with backend APIs using Redux Toolkit, Axios. Implementing features, enhancements, and bug fixes, ensuring code quality and adherence to best practices. Build scripting with shell, managing project dependencies with Yarn, and collaborating with iOS developers to integrate native modules via CocoaPods. Utilizing React Navigation for seamless screen transitions, and Google MUI components for a polished user experience. Integrated Google Maps for static maps, and dynamic routes for logistics feature. Creating adaptive experiences for tablet and mobile, releasing app on internal test channels.

Technologies/Tools: React Native, JSX, React Native Navigation, Redux Toolkit, Axios, Firestore, MUI, Google Maps, Android Studio, Xcode, Gradle, CocoaPods, NPM/Yarn, Expo, Git, GitLab, SQLite, Shell Scripting

Key Achievements:

- Reduced load time by 20% through optimized API calls using Axios and efficient caching with Firestore
- Proposed usage of Google MUI components for ease of adherence to design specifications
- Ensuring best practices for state management using Redux Toolkit
- Google Maps integration using third-party SDK ensured seamless map experience across iOS/Android platforms
- Regularly updating mobile app project with latest packages, with automation, using scripting
- Integrated internal React projects with mobile app project using knowledge of Git, merging JSX views with navigation, which allowed company to accelerate, as integration with internal projects was not successfully accomplished in the past

Senior Fullstack Mobile Application Developer (Full-Time) **ASEBP (From: January 2023, To: December 2023)**

Responsible for the full range of mobile application development, maintenance, and API development. Demonstrated a comprehensive skill set in application development, ensuring seamless user experiences across [iOS/Android](#) platforms. Investigated and developed root cause fixes for data issues across integrated systems. API development comprised of creating well-tested API end-points with ASP.NET Core.

Technologies/Tools: Swift, RxSwift, Core Data, Realm, Java/Kotlin, Xcode, Android Studio, RxJava, SwiftUI/Jetpack Compose, Dagger-Hilt, Visual Studio, ASP.NET Core, Azure

Key Achievements:

- Fullstack role encompassed the full lifecycle of applications, from inception to test planning and maintenance
- Commitment to test-driven development, increased code coverage for iOS/Android applications up to 90% to ensure application robustness using XCTest/JUnit for unit and integration testing
- Utilized App Center, mobile IDEs, for crash logs and profiling to ensure optimal performance on both iOS/Android
- Used a consultative approach with customer groups to gather requirements and design APIs, which reduced rework rate during development, candidate successfully integrated these APIs into the flagship mobile apps
- APIs developed with the ASP.NET Core framework, candidate used Azure to securely setup and configure cloud resources
- Used SQL Server Data Tools with VS 2023 to explore, and update databases, and to create and run SQL scripts
- Candidate developed new features for the flagship mobile apps, proposed and incorporated SDK updates for iOS/Android to favor SwiftUI/Jetpack Compose for seamless app frontend, which saved development time by at most 50%
- For async programming used RxSwift, RxJava to have seamless integration with backend, for iOS used Core Data to manage user JWT sessions, and used SharedPreferences for Android
- Stay-signed-in feature later migrated to use Realm for improved thread safety, and Realm's stateful integrations with SwiftUI/Jetpack Compose

Fullstack Engineer/Cofounder

School2Code (From: February 2022, To: December 2022)

Cofounding role in an ed-tech startup. Worked on deep learning projects using open source technologies such as BERT, Hugging Face Transformers, with PyTorch, and TensorFlow (GPU accelerated). Used DeepPavlov for dialog systems with intents to predefined actions for generating simple coding questions in a PDF format.

- Curating learning modules from core software and game development, UI programming and designing
- Gathered initial interests from alumni groups, understood the potential of language models to generate simple coding questions that combine various aspects of software and game development
- Discussing them with stakeholders, before the wave of modern language models, developing POCs and testing within small groups
- Attending meetups and conferences for understanding technology and customer expectations in the coming years
- **Technologies/Tools:** Docker, ML Kit/CoreML, AWS, Python, React, Webpack, Django, DeepPavlov, PyTorch, TensorFlow

Mobile Application and 3D/Rendering Developer (Full-Time)

Autodesk (From: June 2020, To: January 2022)

Demonstrated meticulous research on cross-platform SDKs, existing solutions at Autodesk for mobile rendering. Contributed to several rendering projects across teams by profiling, evaluating, proposing optimization strategies, and integrating rendering features, while recommending best practices when using VR debugging tools for Quest such as RenderDoc and Qualcomm Profiler. Performed frontend development for single-page web applications with React, seamless integration of 3D assets using Three.js, and harnessed the WebXR API to create cross-device compatible content.

Technologies/Tools: JavaScript, Node.js, React, Webpack, React Expo, Django, Android Studio, Xcode, C#, Unity/Unreal, Fbx/Obj, Revit, Figma

Key Achievements:

- Proficiency in React/WebXR helped to create frontend tools for immersive experiences, which expanded internal customers by 14% on the web front, expertise in Django, mobile app development, Unity, and Revit, further enriched contributions to the prototypes
- Evaluation of async 3D model loaders, such as Expo GO for React, and the glTF runtime loader for Unity/Unreal underscored commitment to cutting-edge technology
- Actively participated in the weekly OpenXR meetings, carried out documentation which contributed to an increased interest (by global teams) to delve into the cross-platform aspects of the XR SDKs
- Explored the boundaries of XR capabilities by meticulously testing and documenting Unity web, standalone builds, and existing solutions at Autodesk for mobile rendering which increased the documentation efforts by rest of the XR team

Mobile Application Developer (Full-Time)

Expedia (From: July 2019, To: May 2020)

Implemented augmented reality features as frameworks for iOS. Hands-on mobile frontend development for iOS/Android for trip [features](#). Delivered APIs with Django and GraphQL. Worked in an agile environment and performed activity planning using Trello and JIRA, while documenting valuable insights on Confluence. To enhance the quality of the apps, used test coverage tools. Notable development of the "[CarryOn-AR](#)" framework - a clever solution for measuring carry-on luggage, published on the App Store.

Technologies/Tools: Xcode, Android Studio, AR/VR, ARKit, MapKit, ARCore, Google Maps, GraphQL/Apollo

Key Achievements:

- Candidate developed and integrated "CarryOn-AR" framework into the main iOS bookings app, and achieved a remarkable 3.1% increase in conversions across various points of sale
- Worked on multi-item-trip project for improved booking experience by allowing multi-destinations, for iOS/Android platforms, both projects were released with feature toggles
- Proficiency with Xcode, Android Studio, ARKit, MapKit, ARCore, Google Maps, allowed to contribute to in-house prototypes which were used for employee training
- Collaborated remotely with cross-functional teams for on-time deliveries

- Met deadlines for quarterly releases by participating in code reviews and adopting architecture recommendations

Fullstack Mobile Application and 3D/Rendering Developer (Full-Time)

Smartivity, Rolocule (From: May 2017, To: July 2019)

Pivotal role in developing the Smartivity Edge app, showcasing expertise in mobile graphics and user experience, while leveraging cutting-edge augmented reality technology. Using technologies such as Node.js, PHP, Android Studio, Xcode, and Unity. Timely launches of the [Smartivity Edge](#) and [Imagenie](#) AR apps across Google Play and App Store. Provided a major update to offload features as downloadable app content, which increased the app performance, reduced the app size by approximately 12%, leading to positive user feedback. At Rolocule, had a dynamic engineering role in a five-person company directly led by the CEO. Creating engaging and interactive gamified applications (for desktop and mobile-native) that blend creativity with technical proficiency for captivating user experiences. Worked in an agile environment. Making necessary code changes to resolve application issues and adding enhancements iteratively to existing applications. Expertise in version control, and teamwork ensured smooth project execution.

Technologies/Tools: Node.js, Java, Android Studio, Swift, Xcode, OpenGL, C#, Unity, AR/VR, MySQL, WinForms, WPF/XAML, NUnit

Key Achievements:

- Created one-of-a-kind input library using miracast and motion sensors, natively on Android
- Implemented software architecture patterns for controller inputs, for seamless UI integration
- Integrated and optimized RPC calls for various in-app actions in a multiplayer environment, which provided network resilience allowing quick recovery of sessions
- Extended internal platform based on OpenGL for rendering models, which improved rendering quality on Windows and provided a rich set of UI controls using SharpGL WPF
- Updated C# plugins for in-app [360 recording](#), which upgraded video quality from SD to HD, with audio channel

Fullstack Software Developer (Full-Time)

MetaDesign Solutions (From: November 2014, To: March 2017)

Contributed to a diverse range of mobile and web applications for clients. Effective requirements gathering to ensure a clear understanding of project scope. Hands-on development, and leadership within small teams, overseeing both backend and mobile frontend development. Technical skills spanning JavaScript, PHP, Unity, and Intel XDK, contributing to successful project outcomes. Cross-platform graphics app development for iOS, Android, Windows platform.

Technologies/Tools: JavaScript, PHP, Unity, MSTest, Intel XDK, iOS/Android

Key Achievements:

- Individual contributor to a multitude of software applications, improving them iteratively each quarter
- Reducing project delivery times for complex projects by integrating testing and tools for automation
- Accomplishing on-time deliveries by thriving in an agile environment, while overseeing mobile and backend projects

Academic Background

Prateek completed his bachelor's in Computer Science from GGSIPU, from 2010 to 2014. He studied subjects such as Software Engineering, Computer Graphics, Applied Mathematics, etc. and passed with 70.15 percentage.

Year	Qualification	University / School	Subjects	CGPA / %
2014	Bachelor of Technology in Computer Science	Indraprastha University (Delhi, India)	Applied Mathematics OOP using C++ & Java Software Engineering Algorithm Analysis & Design Computer Graphics Operating Systems Digital Circuits & Systems Computer Architecture	70.15
2010	Senior Secondary School	St. Gregorios School (Delhi, India)	Physics Chemistry Mathematics Computer Science	75.15

Publications

Prateek is the lead author of the following books. Topics of these books include, but is not limited to, Android, OpenGL ES, Java, Computer Graphics, 3D Rendering, Shaders, JavaScript, Browser Extensions, Software Architecture, Mobile Application Development.

1. [Learn OpenGL ES – For Mobile Game and Graphics Development \(Book\)](#)
2. [Creating Google Chrome Extensions \(Book\)](#)
3. [DataScience \(Blog\)](#)

Open Source

Prateek is keen to contribute to open source projects and forums, and is currently using all the following platforms to participate in the software-dev community. The open source code contribution comprises of packages for frontend development, and sample codes for the mobile, data, backend platforms, with an intent to grow further.

1. Github
 - [EasyThumbs](#), [AsyncImage](#), [SerializedNavPath](#), [SwiftUINavigationController](#) are frontend repos
 - [DockerDjango](#) is a backend repo
2. Bitbucket - [Repos](#) (frontend and backend), [Obj Parser](#) (for OpenGL)
3. Stack Overflow - [Patt Mehta](#)

Certifications

Prateek takes interest in upgrading his skills, understanding new technologies, and getting certified in these by applying the new learnings to early projects that cover use-cases from a broader scope. He has done certifications in data science and geospatial analysis, and applied learnings to create early projects. As such he has certifications in web and backend development, and the following list only covers the recent certifications. The completed projects and certification documents are available on request.

1. Data Scientist/Data Streaming Nano-Degree from Udacity
2. Geographic Information Systems Essentials Professional-Certificate from edX

Lead Software Development

Prateek has demonstrated significant expertise in fullstack software development, platform engineering, and tools engineering. They have also demonstrated abilities of liaising with partners and stakeholders. At Smartivity, Rolocule, they recruited talent by visiting colleges, discussing ongoing projects, and showcasing current work. Their participation in onsite marketing exhibitions at Smartivity allowed them to engage audiences with AR/VR projects. At Autodesk, they facilitated partnerships or participated with industry leaders like NVIDIA, Unity, and OpenXR through recurring meetings and collaborative projects. For e.g., they worked collaboratively on the NVIDIA Omniverse and CloudXR platforms, and the Unity Reflect platform. Their role at School2Code involved gathering stakeholder interests and conducting meetups to discuss product scope and roadmaps. At Mimik Technology, they participated in hackathons and provided an improved outlook to external developer documentations, providing valuable insights to the sitemap.

With the right support, they can take technical ownership for a). *project or subsystem*, b). *architecture decisions*, c). *engineering practices*, and d). *special tools initiatives*. To improve their leadership abilities, they have also recently worked as a part-time instructor to teach fullstack application development certification courses in Vancouver, utilizing tools like VS Code, GitHub, CodeSandbox, and Teams channels.

Their strong communication skills, technical knowledge, and experience in leading diverse groups enabled emerging professionals to create useful software and publish works with confidence on public platforms. With a background as a published author and active contributions on platforms like StackOverflow and GitHub, they have a proven track record of sharing knowledge and fostering collaboration within the developer community. This blend of technical and interpersonal skills ensures they can effectively lead and promote innovative projects with the right support.

AI Developments

As there are growing expectations to integrate AI tools into the engineering workflow, and evaluating AI-generated code, Prateek has also carried out some reformed styles for software development, which can be safely practiced by small teams using local MCP extensions and servers (Claude/OpenAI). These use-case driven approaches for the use of AI for software development have helped to reduce some redundant steps such as code conversions and translations, while ensuring safe, and reliable use of AI. The initiatives undertaken can be discussed upon additional requests.